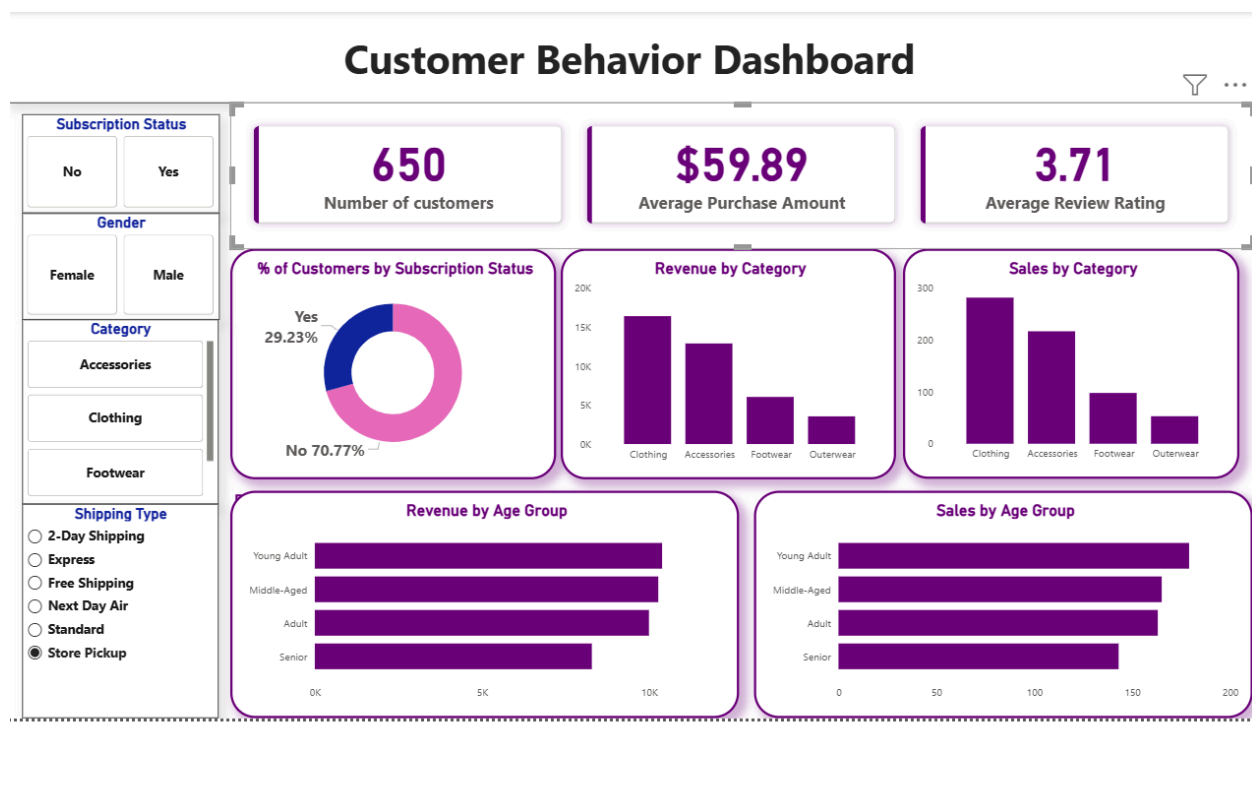


Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.



2. Dataset Summary

- Rows: 3,900
- Columns: 18

Key Features

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

Data Quality

- Missing Data: 37 values in the Review Rating column

```
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

```
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Visa
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	MasterCard
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Visa
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	MasterCard
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Visa

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python.

Data Loading

- Imported the dataset using pandas.

Initial Exploration

- Used `df.info()` to check the dataset structure.
- Used `df.describe()` for summary statistics.

```
[3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Customer ID                          3900 non-null   int64
 1   Age                                  3900 non-null   int64
 2   Gender                              3900 non-null   object
 3   Item Purchased                      3900 non-null   object
 4   Category                            3900 non-null   object
 5   Purchase Amount (USD)               3900 non-null   int64
 6   Location                            3900 non-null   object
 7   Size                                3900 non-null   object
 8   Color                               3900 non-null   object
 9   Season                              3900 non-null   object
10   Review Rating                       3863 non-null   float64
11   Subscription Status                 3900 non-null   object
12   Shipping Type                      3900 non-null   object
13   Discount Applied                   3900 non-null   object
14   Promo Code Used                    3900 non-null   object
15   Previous Purchases                 3900 non-null   int64
16   Payment Method                     3900 non-null   object
17   Frequency of Purchases             3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

Missing Data Handling

- Checked for null values.
- Imputed missing values in the Review Rating column using the median rating of each product category.

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
df.isnull().sum()
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status  0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64
```

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

Column Standardization

- Renamed columns to snake_case format for consistency and readability.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

Feature Engineering

- Created `age_group` column by binning customer ages.
- Created `purchase_frequency_days` column from purchase frequency data.

Data Consistency Check

- Verified whether `discount_applied` and `promo_code_used` were redundant.
- Removed the `promo_code_used` column after validation.

```
df[['discount_applied', 'promo_code_used']].head(10)
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
np.True_
```

```
df = df.drop('promo_code_used', axis=1)
```

```
df.columns
```

Database Integration

- Connected Python to PostgreSQL using SQLAlchemy and Psycopg2.
- Loaded the cleaned dataset into PostgreSQL for SQL analysis.

```
import psycopg2
import sqlalchemy

print(psycopg2.__version__)
print(sqlalchemy.__version__)
```

2.9.12 (dt dec pq3 ext lo64)
2.0.41

```
from sqlalchemy import create_engine

username = "postgres"
password = "root12"
host = "localhost"
port = "5432"
database = "customer_behaviour"

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

table_name = "customer"
df.to_sql(table_name, engine, if_exists = "replace", index=False)

print(f"Connected Successfully! loaded into table `{table_name}` in database `{database}`.")
```

Connected Successfully! loaded into table `customer` in database `customer\_behaviour`.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions.

1. Revenue by Gender

- Compared total revenue generated by male and female customers.

	gender 	revenue 
	text	numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users

- Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
Total rows: 839		Query complete 00:00:00.136

3. Top 5 Products by Rating

- Found products with the highest average review ratings.

	item_purchased text	average_product_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison

- Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers vs. Non-Subscribers

- Compared average spend and total revenue across subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-Dependent Products

- Identified five products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7. Customer Segmentation

- Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 Products per Category

- Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169

9. Repeat Buyers & Subscriptions

- Checked whether customers with more than five purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group

- Calculated total revenue contribution of each age group.

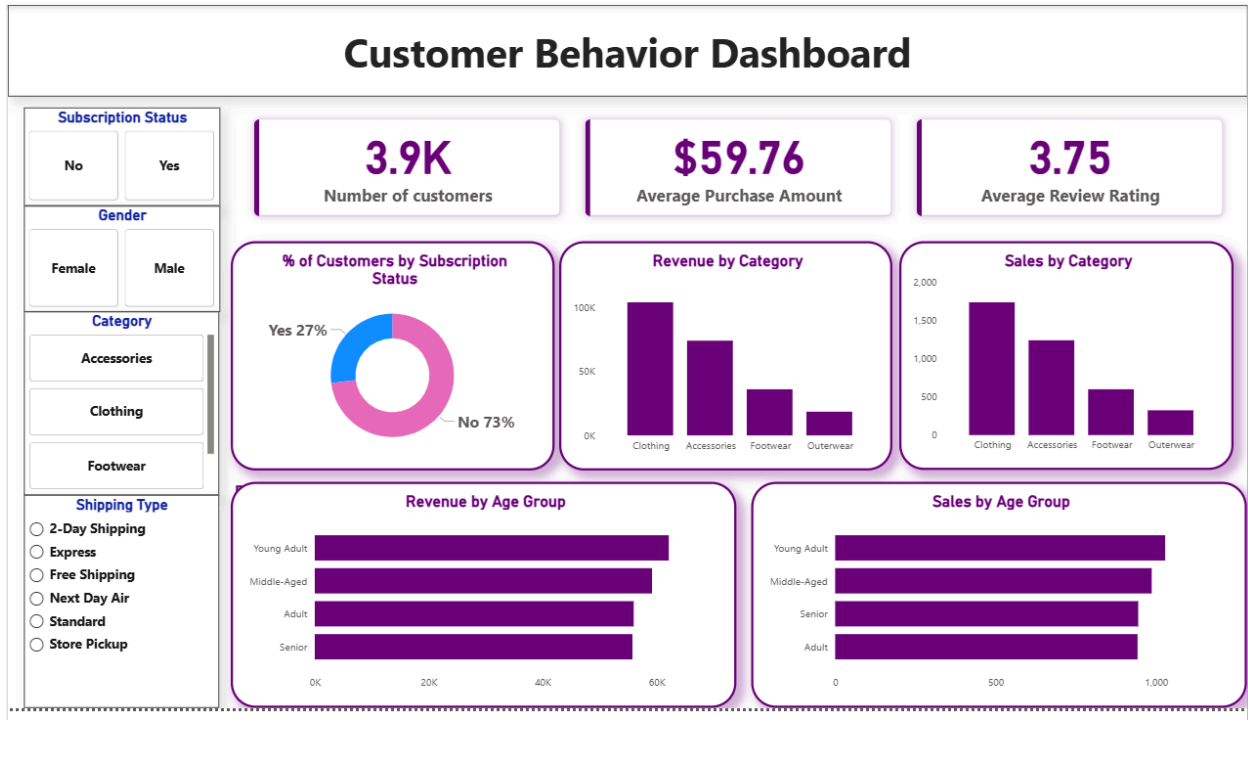
	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-Aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, an interactive dashboard was built in Power BI to visualize customer behavior, revenue trends, product performance, and customer segments.

Dashboard Features

- Revenue KPI
- Customer Count KPI
- Revenue by Gender
- Revenue by Age Group
- Subscription Analysis
- Customer Segmentation
- Product Performance Analysis
- Discount Analysis



6. Business Recommendations

Boost Subscriptions

- Promote exclusive benefits and personalized offers for subscribers.
- Encourage repeat customers to join the subscription program.

Customer Loyalty Programs

- Reward repeat buyers and provide incentives to move customers into the Loyal segment.

Review Discount Policy

- Balance the positive impact of discounts on sales with profit margin considerations.

Product Positioning

- Highlight top-rated and best-selling products in marketing campaigns and promotional activities.

Targeted Marketing

- Focus marketing efforts on high-revenue age groups and customers with higher purchase frequency.
-

7. Conclusion

This project demonstrates how Python, PostgreSQL, and Power BI can be integrated to analyze customer shopping behavior and generate actionable business insights. The findings support data-driven decision-making and help businesses improve customer engagement, retention, and overall sales performance.

8. GitHub Repository

GitHub Repository Link:

<https://github.com/MaheshaVP/Customer-Shopping-Behavior-Analysis>